Reward Shaping Is Noise Shaping

A short brief: the idea, the diagnosis, the method, and where it stands

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The whole project in five sentences

1. In RL with verifiable rewards (RLVR), many methods change how answers are scored (reward right ones, penalize wrong ones, change the baseline, aim at pass@k) and claim more *diversity*.
2. They are compared at the same training step; a method that learns more slowly is less sharpened at that step, so it *looks* more diverse.
3. We prove that every correctness-only scoring rule pushes the model the same way on average and differs only in noise; noise sets the *speed* along one accuracy–diversity curve, not the curve.
4. Experiments on two model families confirm it, and adding noise alone recreates the effect of the best-known method (NSR).
5. Diversity at equal accuracy is only how evenly success is spread across questions, so a real method must bring new signal to rarely solved questions; we are now testing one that does.

1 The words you need

pass@1	Accuracy: the chance one sampled answer is right.
pass@k	The chance at least one of k answers is right: the usual measure of "diversity" or coverage.
Update	$g = \sum_i A_i \nabla \log \pi(y_i)$: each answer y_i is pushed up or down by its advantage A_i.
Dr. GRPO	$A = r - \bar{r}$ (reward minus the group's mean); the standard, low-noise baseline.
PSR / NSR	Reward right answers only ($A = r$) / penalize wrong answers only ($A = r - 1$). NSR is claimed to preserve diversity.
Matched accuracy	Compare pass@16 *at the same pass@1*, by interpolating along the reference run's training curve.

2 The theory

For any rule that scores answers from their correctness, with p a question's success rate:

$$\mathbb{E}g = c(p)\,\nabla p, \qquad \mathbb{E}\|g\|^2 = F_1 D_1(p) + F_0 D_0(p) + \|\nabla p\|^2 X(p),$$

where F_1 and F_0 are the mean squared gradients of a right and a wrong answer. The *direction* is always ∇p; a rule can only reweight questions ($c(p)$) and change the noise (D_1, D_0). PSR and NSR

have the same mean; NSR is PSR plus zero-mean noise, carried by the wrong answers, which have 2–3× larger squared gradients (measured on both families).

A useful identity. At matched pass@1, pass@$k = \mathbb{E}_x[1 - (1 - p_x)^k]$ is concave in each question's p_x, so it is highest when success is spread evenly across questions. Diversity at equal accuracy measures that spread, nothing else.

3 The diagnosis (finished)

Four findings

1. **One curve.** On Qwen2.5-Math-1.5B, PSR, NSR, five baselines in between, the NSR paper's own implementation and a pass@4 objective all move along one accuracy–diversity curve. On OLMo-2-1B: NSR minus PSR at matched accuracy −0.30, 90% [−1.28, +0.77], **equivalent within** ±2; NSR is 19–22× noisier.
2. **Noise is the cause.** Adding to PSR a random-sign term of NSR's size (same mean) makes it exactly as slow as NSR: speed $\phi = -0.11$ [−0.22, −0.05] on a scale where NSR = 0 and PSR = 1. NSR at twice the learning rate recovers its lag. All stay on one curve (3 fresh seeds).
3. **The theory's noise formula fits** training within 0.69–1.18 once gradient sizes are re-measured; the right/wrong asymmetry fades from about 3× to 1.5× during training. With Adam, noise does not shrink steps; it misdirects them (13–27% less aligned).
4. **Not a temperature effect.** RL barely changes entropy on fixed text (1.081 → 1.064) but cuts it 3–4× on its own samples: it steers into confident paths. Sampling a late checkpoint hotter does not beat the training curve (−1.42, 90% [−2.19, −0.68], 511 problems).

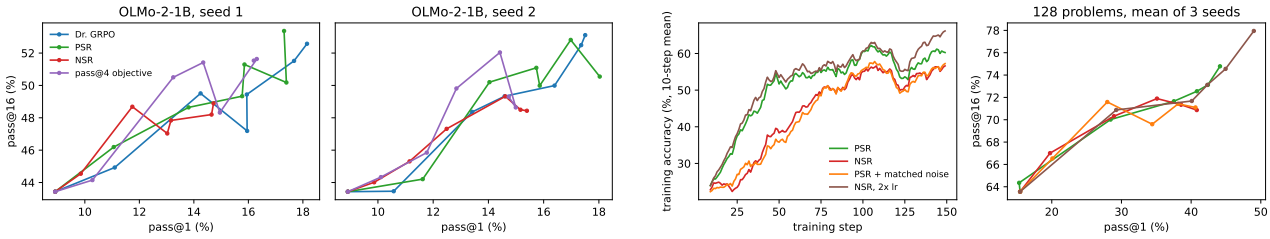


Figure 1: Left: OLMo, five rules on one accuracy–diversity curve. Right: PSR plus matched noise (orange) learns as slowly as NSR (red); all lie on one curve.

Independent review (Codex as an ICML reviewer): 4/10 at first (claims too broad, cause not shown) → 5/10 after the noise test → **6/10** after the OLMo replication. To reach 7/10 the paper needs a method, derived from the diagnosis, that moves the curve.

4 What the diagnosis says a method must do

Finding	Requirement
Correctness-only rules share one curve	Bring information the current samples do not contain
Diversity = spread of success across questions	Raise success on rarely solved questions
On those questions a group of 8 answers is usually all wrong	Supply signal exactly there
No global sharpening; entropy control blocks learning	Do not regulate entropy
Extra noise slows learning	Keep Dr. GRPO's low-noise update as the base

5 Method, version 1: coverage-weighted memory (Pre-reg. 12)

Coverage-weighted memory (CM)

Train with Dr. GRPO. Store the verified, completed correct answers the model finds (at most 4 per question). Each step, replay 8 stored questions with an extra gradient $\lambda \nabla \log \sum_{y \in S_q} \pi(y \mid q)$, choosing questions in proportion to $k\,p(1-p)^{k-1}$ ($k = 16$). That weight is exactly $\partial\,\text{pass@}k / \partial \log p$: replay goes to questions solved about once in k, whose pass@k can still rise.

Pilot (1 seed, 255 validation problems): memory was the only family that lifted the curve (coverage memory +2.1; uniform memory +0.2); entropy bonus and the unlikeliness reward did nothing; entropy floors and adaptive entropy collapsed learning.

Main runs (3 fresh seeds, 511 test problems with audited answer keys). Pre-registered rule: *moves the curve* if at least +2 points of pass@16 above Dr. GRPO's curve, including its best temperature, with the 95% interval above 0, in all three seeds. **Final result (October 21):**

Arm (seeds 8, 9, 10)	Gain at matched accuracy	95% interval	Step-150 pass@1 / greedy test acc.
Dr. GRPO (reference)	0		44.1 / 47.9
Coverage memory	+1.19 (per seed +1.5, +1.4, +0.6)	[+0.54, +1.79]	42.1 / 47.7
Uniform memory	+0.61	[−0.04, +1.27]	42.9 / 47.3
Unlikeliness reward (published)	+0.14	[−0.32, +0.62]	43.4 / 47.6

Registered verdict: "average agreement". A small, consistent lift (positive in every seed, interval above 0, larger than uniform replay and the published unlikeliness reward), but below the +2 bar; coverage weighting over uniform replay (+0.66) misses its +1 bar. Greedy test accuracy differs by −0.18 points, 90% [−0.83, +0.43]; sampled pass@1 at step 150 is 2 points lower. **Why so small:** in 150 steps each training question is visited about once; a group of 8 finds a success on a question with $p = 1/16$ only 40% of the time ($p = 1/32$: 22%), so the memory misses most of the questions it is meant to help.

6 Method, version 2: seed the memory from the base model (Pre-reg. 14, running)

Before training, sample the base model 32 times on every training question ("discovery"); this finds a success on a $p = 1/16$ question 87% of the time (for $p = 1/32$: 64%). Seed the memory with one base success per question; replace it once the model finds its own. Each arm answers one question about *why* it works:

Arm (3 seeds each)	Question it answers
D32: seeded coverage memory (candidate)	Does the method move the curve?
U32: seeded, uniform replay	Does the coverage weighting matter?
S32: fine-tune on the same successes, then plain RL	Does continual replay beat a one-off warm start?
C05: unseeded coverage memory, same strength	Does the seeding matter?
G32: plain Dr. GRPO, groups of 32	Is it just "more samples"?

Pre-registered win: D32 at least +2 points above Dr. GRPO's curve and temperature envelope (95% interval above 0, all three seeds) *and* equal final accuracy (90% interval within ±1); each mechanism claim needs D32 $\geq +1$ over that arm's curve. All six tests are corrected together (Holm). Declared and reviewed by Codex before any training; amendments recorded before outcomes.

Status on October 21, 2026

- Smoke test passed: all arms run, the memory is seeded, base traces retire as the model finds its own successes, memory use equals Dr. GRPO's.
- Discovery done (125,440 base-model answers): 3,433 of 3,920 training questions solved at least once; 3,235 with a retained success, including 988 of the 1,181 rarely solved ones (1–4 successes in 32).
- Training launched for D32, U32, S32, C05 (one 4-GPU job per seed) and G32; waiting for GPUs. About 3.5 h of training (G32 about 9 h), then about 4 h of evaluation.

7 What each outcome means for the paper

Outcome	Paper
D32 wins, and beats U32, S32, G32	Diagnosis + a method derived from it (the 7/10 target); next, replicate on OLMo.
D32 wins but ties a control	The method works; the mechanism claim is scoped (e.g. "more samples help").
No win	A strong diagnostic paper: popular diversity fixes (NSR, entropy control, unlikeliness, pass@k objectives) do not move the curve, the theory says why, and replay of rare successes is the only lever that moved it (about 6/10).

Every experiment was pre-registered and reviewed before launch; the full story, with all twelve experiments, is in `report/story.pdf`.